

SpaceExplorer · Student worksheets

Ages 8–12 About 10 min · per mission

Paper, a pencil and these pages. A connected device is optional.

Mission 1 · About 10 min

One day, one year

Two movements, two clocks

Learning goals

Tell the difference between turning on an axis and travelling around the Sun.

Read and discover

Earth turns on its axis, an imaginary line through its poles. Your part of Earth moves into sunlight and then away from it. This gives us day and night.

Earth also travels around the Sun. One complete trip is a year: about 365 Earth days. A day from one noon to the next lasts about 24 hours.

Mars travels around the Sun too, but its year lasts about 687 Earth days. Turning on an axis is rotation; travelling around another object is an orbit.

Activity: Try the two clocks

With the screen: move the time slider to about 365 Earth days, then 687. Compare Earth's and Mars's progress around the Sun. Read each planet's orbital time.

On paper, draw a turning Earth and its path around the Sun. Which movement explains night? Which planet takes longer to finish one orbit?

My observations

Diagrams and the 3D view simplify sizes and distances. Compare the numbers in kilometres to check real sizes.

Open the lesson: <https://sadik-dev.github.io/SpaceExplorer/learn/?lang=en&mission=day-year>

Sources: NASA · Mars facts - <https://science.nasa.gov/mars/facts/>

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Mission 1

One day, one year

Questions

Choose one answer.

1. What makes day and night on Earth?

- ☐ The Sun switches off
- ☐ Earth turns on its axis
- ☐ The Moon covers Earth every night

2. What is one year on a planet?

- ☐ One complete orbit around the Sun
- ☐ One turn on its axis
- ☐ One trip by its moon

3. Earth's year is about 365 days; Mars's is about 687. Which is longer?

- ☐ Earth's year
- ☐ They are equal
- ☐ Mars's year

One last thought

If you lived on Mars, would birthdays counted in local years come more or less often? Explain why.

Open the lesson: <https://sadik-dev.github.io/SpaceExplorer/learn/?lang=en&mission=day-year>

Sources: NASA · Mars facts - <https://science.nasa.gov/mars/facts/>

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Mission 2 · About 10 min

Meet the planet families

Rocky worlds and giant neighbours

Learning goals

Sort the eight planets into three families and use measurements to compare their sizes.

Read and discover

Mercury, Venus, Earth and Mars are rocky planets. They have solid surfaces. These four worlds are the closest planets to the Sun.

Jupiter and Saturn are gas giants. Uranus and Neptune are ice giants. The giants have deep atmospheres and no solid surface like Earth's. "Ice giant" does not mean a frozen ball you could stand on.

Diameter measures across a world through its centre. In our atlas, Earth's mean diameter is 12,742 km and Jupiter's is 139,822 km: roughly eleven Earth diameters. Picture sizes alone can mislead you.

Activity: Become a planet sorter

With the screen: choose Earth and Jupiter in the comparison. Read their diameters. Then find Uranus and its family.

Make three headings on paper: rocky, gas giant and ice giant. Put all eight planet names under the right heading. Circle the larger diameter: 12,742 km or 139,822 km.

My observations

Diagrams and the 3D view simplify sizes and distances. Compare the numbers in kilometres to check real sizes.

Open the lesson: <https://sadik-dev.github.io/SpaceExplorer/learn/?lang=en&mission=planet-families>

Sources: NASA · The planets - <https://science.nasa.gov/solar-system/planets/>

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Mission 2

Meet the planet families

Questions

Choose one answer.

1. Which of these is a rocky planet?

- ☐ Mars
- ☐ Jupiter
- ☐ Neptune

2. Which pair belongs to the ice giant family?

- ☐ Earth and Mars
- ☐ Jupiter and Saturn
- ☐ Uranus and Neptune

3. How can you reliably compare Earth's and Jupiter's real sizes here?

- ☐ Count their moons
- ☐ Compare their diameters in kilometres
- ☐ Compare only their picture sizes

One last thought

Why might a diagram make a small planet look larger than it should? What number could you check?

Open the lesson: <https://sadik-dev.github.io/SpaceExplorer/learn/?lang=en&mission=planet-families>

Sources: NASA · The planets - <https://science.nasa.gov/solar-system/planets/>

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Mission 3 · About 10 min

Our changing Moon

One Moon, many views

Learning goals

Explain moonlight, phases and the Moon's journey around Earth.

Read and discover

The Moon shines in our sky by reflecting sunlight. It does not make its own visible light like the Sun. Normally, the Sun lights half of the Moon.

The Moon orbits Earth while Earth and the Moon travel around the Sun together. From Earth, we see different amounts of the Moon's sunlit half as it moves. These views are called phases.

At full Moon, we see the bright half facing us; at new Moon, the bright half faces away. The phase cycle lasts about 29.5 days. Earth's shadow causes a lunar eclipse, not the regular phases.

Activity: Change your view

With the screen: move through the four example phases. Watch how much of the bright part is visible from Earth. The diagram shows a simplified view, not tonight's Moon.

Draw a new Moon, a quarter Moon and a full Moon as seen from Earth. Label where the light comes from. Does the Moon itself change shape?

My observations

Diagrams and the 3D view simplify sizes and distances. Compare the numbers in kilometres to check real sizes.

Open the lesson: <https://sadik-dev.github.io/SpaceExplorer/learn/?lang=en&mission=earth-moon>

Sources: NASA · Moon phases - <https://science.nasa.gov/moon/moon-phases/>

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Mission 3

Our changing Moon

Questions

Choose one answer.

1. Where does moonlight come from?

- ☐ Lamps on the Moon
- ☐ The Moon burns like a star
- ☐ Sunlight reflected by the Moon

2. Why do we see different Moon phases?

- ☐ We see different amounts of its sunlit half
- ☐ Earth's shadow covers it every month
- ☐ The Moon changes its actual shape

3. Which path does the Moon follow?

- ☐ It stays still beside Earth
- ☐ It orbits Earth as both travel around the Sun
- ☐ It orbits Mars

One last thought

A friend says the Moon becomes a banana shape. How would you explain what really changes?

Open the lesson: <https://sadik-dev.github.io/SpaceExplorer/learn/?lang=en&mission=earth-moon>

Sources: NASA · Moon phases - <https://science.nasa.gov/moon/moon-phases/>

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